Global warming and choices

Today I can… ☐ I can put cause before effect. ☐ I can say how gases add to warming. ☐ I can compare a cut in emissions with help for an impact.

Lesson strips

Science · GROW · Autumn 2 · Week 4 · SCI-GROW-A2-W04 · Date _____

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Arrival · Supported

Sort these: weather today; climate over many years.

Point to the long-term idea.

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Exit ticket · Supported

Name a cause, an effect and an action. Which link does your action change?

Read it back: explain a cause and an effect of global warming, then compare two kinds of action.

My feedback was… say what changed. Next I need to… name a fix.

How I know: I can put cause before effect. I can say how gases add to warming. I can compare a cut in emissions with help for an impact.

The adult hears you and chooses our next example. Take one quiet minute.

My confidence: □ 1 □ 2 □ 3

I · S · C □

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Arrival · Independent

Sort these: weather today; climate over many years.

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Exit ticket · Independent

Name a cause, an effect and an action. Which link does your action change?

Read it back: explain a cause and an effect of global warming, then compare two kinds of action.

My feedback was… say what changed. Next I need to… name a fix.

How I know: I can put cause before effect. I can say how gases add to warming. I can compare a cut in emissions with help for an impact.

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Arrival · Aspire

Sort these: weather today; climate over many years.

Point to the long-term idea.

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Exit ticket · Aspire

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Read it back: explain a cause and an effect of global warming, then compare two kinds of action.

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Starter

Think back. Say, point or write one answer for each question.

1. Name one fact about the Solar System and its class card.

2. Why does one place have night?

3. Name one way friction can affect motion.

Today I will check:

I can put cause before effect.

I can say how gases add to warming.

I can compare a cut in emissions with help for an impact.

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Today I can… ☐ I can put cause before effect. ☐ I can say how gases add to warming. ☐ I can compare a cut in emissions with help for an impact.

I do

I order the chain: burn fuel → more carbon dioxide → more heat stays → warming.

Cutting emissions acts near the start. A flood defence helps with an impact.

Watch: Burn fossil fuel → add carbon dioxide → more heat stays → heat risks. Cut fuel use to cut emissions; plan shade to help with heat.

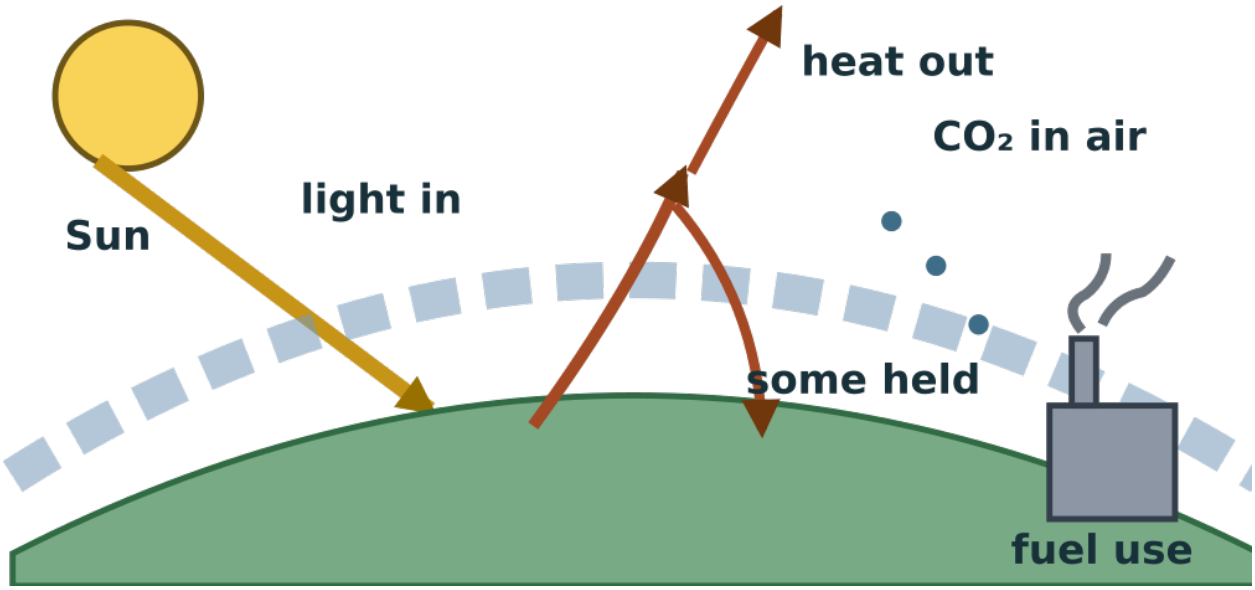
Show me: Point to the cause card: burning fuel or a hotter day?

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Global warming and choices

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Warming mechanism



Light comes in. Earth releases heat. More CO_2 slows heat loss. Simplified mechanism; not to scale.

Picture colours: yellow sunlight arrow, red outgoing heat arrows, blue air, dark fuel source.

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Global warming and choices

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We do

The burn fuel and impact cards are in place. Sort the missing gas and warming cards. Explain the link.

Our turn: Burn fuel → add carbon dioxide → ____ → heat risks.

Talk, then sort.

Burn fossil fuel. Choose cause, middle link, impact or action.

More greenhouse gas. Choose cause, middle link, impact or action.

Heat risk in some places. Choose cause, middle link, impact or action.

Cut fuel use. Choose cause, middle link, impact or action.

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We do support

Supported: Lay the first and last cards; point to each missing space before choosing a card.

Independent: Explain two chain links.

Aspire: Explain why a flood defence does not remove carbon dioxide already added by burning fuel.

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Ice observation

Guess first: Will ice on dark and light card melt alike?

Kit: Two equal ice pieces. Two trays. Dark card. Light card. Staff clock.

1. Put equal ice pieces on cards in trays.
2. Place both trays in the same light.
3. Observe both at one set time.
4. Record what you see. Say why this is not a climate model.

Risk: Melted ice can leave a slippery spill.

Safe: Keep ice in trays. Wipe spills and use no heater.

This is a surface observation. It is not proof of the greenhouse effect.

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Ice observation record

Time	dark card ice	light card ice
Start	observation	observation
Later	observation	observation

Copy this table into your book. Record what you see at Start and Later.

Keep the ice pieces equal at the start. Use the same light and one set time for both.

Record the actual result. It is not a climate model or proof of the greenhouse effect.

Keep ice in trays. Wipe spills. Use no heater or heat source.

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Global warming and choices

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We do 2

Choose what follows burning more fossil fuel in our warming chain. Everyone answers together.

Which action helps with an impact rather than cutting emissions?

Flood defence

Cut fuel use

All of us: A bus depot burns more diesel. Which immediate link belongs in its emissions chain?

A. More carbon dioxide is released.

B. The extra carbon dioxide appears only when the depot has less daylight.

C. The depot warms first; that warmth makes its engines release carbon dioxide.

D. Only local air pollution changes; the amount of carbon dioxide stays the same.

Point. Tell a partner why.

Take one quiet minute to reset.

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Global warming and choices

Today I can… □ I can put cause before effect. □ I can say how gases add to warming. □ I can compare a cut in emissions with help for an impact.

You do · Supported

Build a cause-to-impact chain. Compare an emissions cut with help for an impact. Give one reason for each.

Talk first. You can plan in any language.

Point, sign, speak, draw, type, write, use a scribe or show a staff-captured photo.

Order one cause and effect.

Say first: First ___, then ___.

I · S · C □

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Global warming and choices

Today I can… □ I can put cause before effect. □ I can say how gases add to warming. □ I can compare a cut in emissions with help for an impact.

You do · Independent

Build a cause-to-impact chain. Compare an emissions cut with help for an impact. Give one reason for each.

Talk first. You can plan in any language.

Point, sign, speak, draw, type, write, use a scribe or show a staff-captured photo.

Explain two chain links.

Say first: Burning ___ adds ___; this leads to ___ because ___.

I · S · C □

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Global warming and choices

Today I can… □ I can put cause before effect. □ I can say how gases add to warming. □ I can compare a cut in emissions with help for an impact.

You do · Aspire

Build a cause-to-impact chain. Compare an emissions cut with help for an impact. Give one reason for each.

Talk first. You can plan in any language.

Point, sign, speak, draw, type, write, use a scribe or show a staff-captured photo.

Compare an emissions cut with an adaptation.

Say first: Action A changes ___, while action B helps with ___; I would choose ___ because ___.

I · S · C □

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Global warming and choices

Today I can… ☐ I can put cause before effect. ☐ I can say how gases add to warming. ☐ I can compare a cut in emissions with help for an impact.

Go deeper

Build a cause-to-impact chain. Compare an emissions cut with help for an impact. Give one reason for each.

Talk first. You can plan in any language.

Point, sign, speak, draw, type, write, use a scribe or show a staff-captured photo.

Why might a council cut fuel use and plan flood defences? Name a limit of each.

Two kinds of action: Choose a council goal, justify the first action and explain the problem still left.

I · S · C ☐

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Global warming and choices

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Review and improve

Hear feedback. Move one chain card and say why.

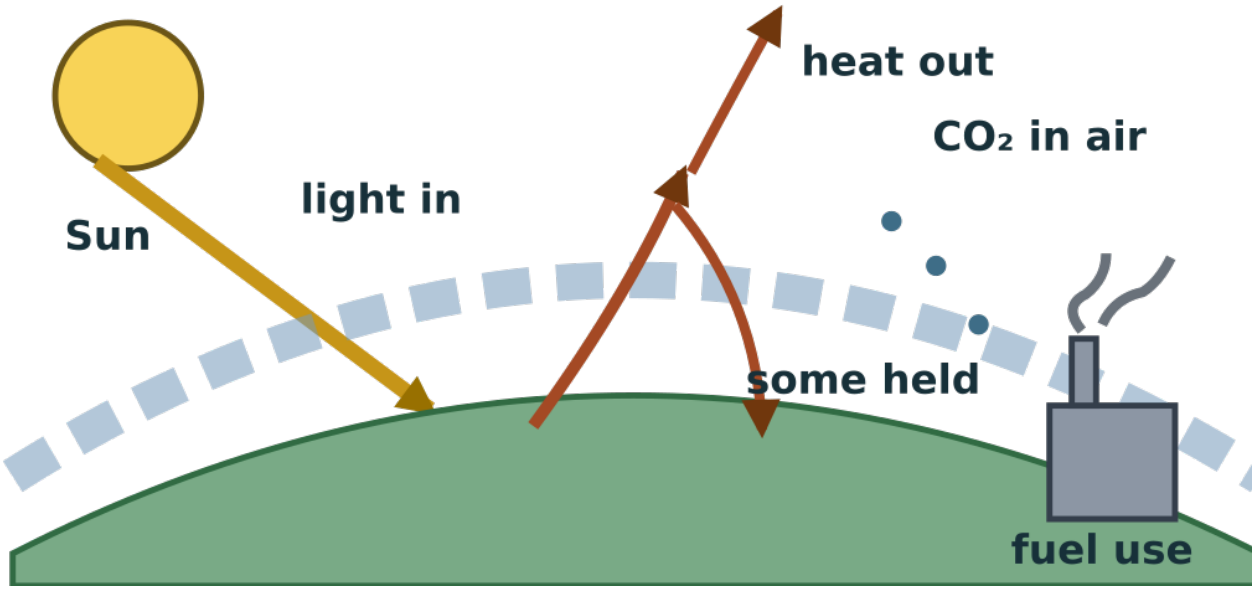
VF: hear one comment. NS(+): name a next step. //: edit one part. ?: ask for a read-back. Take one quiet minute.

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Global warming and choices

Today I can… ☐ I can put cause before effect. ☐ I can say how gases add to warming. ☐ I can compare a cut in emissions with help for an impact.

Knowledge organiser



I am learning to… explain a cause and an effect of global warming, then compare two kinds of action.

Light comes in. Earth releases heat. More CO_2 slows heat loss. Simplified mechanism; not to scale.

Cutting emissions acts near the start. A flood defence helps with an impact.

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Global warming and choices

Today I can… ☐ I can put cause before effect. ☐ I can say how gases add to warming. ☐ I can compare a cut in emissions with help for an impact.

Knowledge organiser continued

climate: the pattern of weather over many years

carbon dioxide: a gas added to the air when fossil fuels burn

greenhouse gas: a gas in the air that slows heat leaving Earth

impact: an effect on people or places

Say first: Burning ___ adds ___; this leads to ___ because ___.

You can plan in any language.

I can put cause before effect.

I can say how gases add to warming.

I can compare a cut in emissions with help for an impact.

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Global warming and choices

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Word page

climate

the pattern of weather over many years

Climate is not the weather on one day.

carbon dioxide

a gas added to the air when fossil fuels burn

Burning coal adds carbon dioxide to the air.

greenhouse gas

a gas in the air that slows heat leaving Earth

Carbon dioxide is a greenhouse gas.

impact

an effect on people or places

More heat can have an impact on people.

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